



Module 1 – Descriptive Statistics and Distributions

Part A – Central Tendency and Spread

1. Marks: 50, 60, 65, 70, 75, 80, 85, 90
 - Find mean, median, mode.
 - Replace 90 with 900 and recompute mean. Explain the effect of outliers.
2. Temperatures (°C): 29, 31, 33, 33, 32, 31, 30
 - Calculate variance and standard deviation.
 - Which measure is easier to interpret and why?

Part B – Percentiles, IQR and Z-Score

3. Cat weights (kg): 2.5, 3.0, 3.2, 3.3, 3.4, 3.5, 3.6, 3.9, 4.0, 4.5
 - Find P25, P50, P75 and IQR. Mark possible outliers.
4. Given $\mu = 50$, $\sigma = 10$
 - Find Z for $x = 65$ and interpret.
 - For $Z = -2.5$, find x .



Module 2 – Probability Basics for ML

Part A – Basic Probability

1. In a survey of 100 people:
 - 40 like pizza (A)
 - 50 like burgers (B)
 - 20 like both ($A \cap B$)

Find $P(A)$, $P(B)$, $P(A \cap B)$, and $P(A | B)$.

Part B – Conditional Probability and Bayes

2. Email dataset of 1 000 mails:

- 100 spam (S)
 - 40 of these contain “free” (F)
 - 60 non-spam also contain “free”
- Compute $P(S | F)$ using Bayes’ Theorem →

Explain what this means for a spam filter.

3. Disease example:

- $P(\text{Disease}) = 1 / 10\,000$
- $P(+ | D) = 0.99$
- $P(- | \neg D) = 0.99$

Find $P(D | +)$ and discuss the base-rate effect.

Part C – Confusion Matrix and Performance Metrics

Task 1:

A binary classifier results:

	Predicted Positive	Predicted Negative
Actual Positive	120	30
Actual Negative	80	770

Compute:

- $\text{Accuracy} = (TP + TN) / \text{Total}$
 - $\text{Precision} = TP / (TP + FP)$
 - $\text{Recall} = TP / (TP + FN)$
 - $F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$
 - $\text{Specificity} = TN / (TN + FP)$
 - $\text{NPV} = TN / (TN + FN)$
 - $\text{Prevalence} = (\text{Actual Positive}) / \text{Total}$
- Explain what each metric tells you about model behavior.

Task 2:

A dataset of 10000 samples has only 2 % positives. The model predicts everything as negative.

- Write the confusion matrix.
- Compute Accuracy, Precision, Recall, F1, Specificity.
- Why is Accuracy misleading in this case?
- Which metric would be more appropriate for imbalanced data and why?

Hospital Triage Model [Optional]

A hospital triage model predicts the condition of patients as:

U = Urgent, N = Normal, and R = Review (within 24 hours).

On a test set of **1,200 patients**, the results are:

Actual \ Predicted	U	N	R
U	120	42	18
N	60	612	48
R	45	90	165

Tasks

① For each class (**U**, **N**, and **R**) — treat it as “positive” (one-vs-rest) — compute:

- Precision (PPV)
- Recall
- F1 Score
- NPV

② Compute **overall Accuracy** of the model.

③ Combine **U** and **R** as one group of “Positive” (needs attention) and **N** as “Negative.”
Then compute:

- Accuracy
- Precision (PPV)
- Recall
- F1 Score
- NPV

④ Briefly interpret which class or group performs best and why this matters in a medical triage context.

